WORKED SOLUTION

STEP III 2024, Question 8

Pure •

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

A quadratic expression of the form $y^2 - x^2 + \dots$ splits into two lines of gradients ± 1 exactly when completing the square leaves no constant behind. The trick in the rest of the question is that adding a multiple of one curve's equation to the other's doesn't lose any common points, and a well-chosen multiple turns the messy quartic problem into "which lines meet a parabola", which is just two quadratics.

Solution

Part (i)

A product is zero exactly when one of its factors is zero, so the points satisfying $(y - x + 3)(y + x - 5) = 0$ are those on $y = x - 3$ (gradient 1) together with those on $y = -x + 5$ (gradient -1).

For the general equation, complete the square in both variables:

$$y^2 - x^2 + py + qx + r = (y + \tfrac{1}{2}p)^2 - (x - \tfrac{1}{2}q)^2 - K, \qquad \text{where } K = \tfrac{1}{4}(p^2 - q^2) - r.$$

Write $Y = y + \frac{1}{2}p$ and $X = x - \frac{1}{2}q$, a translation, which doesn't change gradients. The equation becomes $(Y - X)(Y + X) = K$.

If $p^2 - q^2 = 4r$, then $K = 0$ and the equation is $(Y - X)(Y + X) = 0$, which by the first paragraph is the pair of lines $Y = X$ and $Y = -X$, with gradients 1 and -1. In the original coordinates these are

$$y - x + \tfrac{1}{2}(p + q) = 0 \quad \text{and} \quad y + x + \tfrac{1}{2}(p - q) = 0.$$

Conversely, suppose the equation represents a pair of lines with gradients ± 1; we show $K = 0$. One of the lines has gradient 1, so has the form $Y = X + c$, and every point of it satisfies the equation. On it, $(Y - X)(Y + X) = c(2X + c)$, so $2cX + c^2 = K$ for *every* real X. A linear function of X is constant only if its coefficient vanishes, so $c = 0$, and then $K = c^2 = 0$. Hence $p^2 - q^2 = 4r$, and the "if and only if" is proved.

Part (ii)

If a point lies on C_1, then $y^2 + 2sy + s(s+1) - x = 0$ there; if it lies on C_2, then $y - x^2 = 0$ there. So at a common point the expression

$$y^2 + 2sy + s(s+1) - x + k(y - x^2)$$

is $0 + k \cdot 0 = 0$, whatever the value of k.

With $k = 1$ the equation is

$$y^2 - x^2 + (2s+1)y - x + s(s+1) = 0,$$

which has the form of part (i) with $p = 2s+1$, $q = -1$, $r = s(s+1)$. Then

$$p^2 - q^2 = 4s^2 + 4s + 1 - 1 = 4s(s+1) = 4r,$$

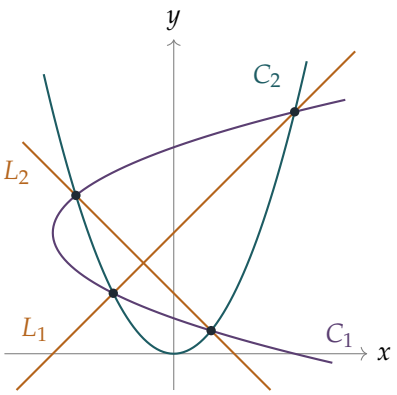
so by part (i) this is a pair of straight lines with gradients 1 and -1. Using the factorisation from part (i), with $\frac{1}{2}(p+q) = s$ and $\frac{1}{2}(p-q) = s+1$, the equation is

$$(y - x + s)(y + x + s + 1) = 0,$$

so the lines are

$$L_1 : \ y = x - s \qquad \text{and} \qquad L_2 : \ y = -x - s - 1.$$

By the first paragraph every point of $C_1 \cap C_2$ satisfies this equation, so all four intersection points lie on $L_1 \cup L_2$. (This holds for every s; we never needed the four points to exist.)



The figure shows the case $s = -2$: C_1 is $x = (y-2)^2 - 2$, and the four common points of C_1 and C_2 lie two on each of the lines $y = x + 2$ and $y = 1 - x$.

Part (iii)

Every common point of C_1 and C_2 lies on C_2 and on L_1 or L_2. Substituting $y = x^2$:

$$\begin{aligned} &\text{on } L_1 : \quad x^2 - x + s = 0, \\ &\text{on } L_2 : \quad x^2 + x + s + 1 = 0. \end{aligned}$$

A point on C_2 is fixed by its x-coordinate, so each line meets C_2 in at most two points. Four distinct common points therefore force each quadratic to have two distinct real roots, so both discriminants are positive:

$$1 - 4s > 0 \quad \text{and} \quad 1 - 4(s+1) > 0.$$

The second gives $s < -\frac{3}{4}$, as required. (The first, $s < \frac{1}{4}$, is then automatic.)

Ask yourself In part (iv), why isn't it enough to find four points on C_2 and $L_1 \cup L_2$?

Part (ii) only showed that common points of C_1 and C_2 lie on the lines; we need the reverse direction too, and we must make sure the four points don't coincide.

Part (iv)

Suppose $s < -\frac{3}{4}$. Then both discriminants above are positive ($-3 - 4s > 0$, and $1 - 4s > 4 > 0$), so L_1 meets C_2 at two distinct points and so does L_2.

These four points are distinct. A point counted twice would lie on both L_1 and L_2. The lines meet where $x - s = -x - s - 1$, that is, at $(-\frac{1}{2}, -\frac{1}{2} - s)$. This lies on C_2 only if $-\frac{1}{2} - s = \frac{1}{4}$, that is, $s = -\frac{3}{4}$, which is excluded. So we have four distinct points.

Each of them lies on C_1. At each point $y - x^2 = 0$ (it is on C_2), and the $k = 1$ equation of part (ii) holds (it is on L_1 or L_2). Subtracting,

$$y^2 + 2sy + s(s+1) - x = 0,$$

which is the equation of C_1.

So C_1 and C_2 have at least four distinct common points; by part (iii)'s argument they can't have more than four. Hence they intersect at exactly four distinct points.

Checks

- Substituting $y = x^2$ into C_1 gives the quartic $x^4 + 2sx^2 - x + s^2 + s = 0$. Multiplying out $(x^2 - x + s)(x^2 + x + s + 1)$ gives exactly this, which confirms the pair of lines.
- With $s = -2$ the four points are $(2, 4)$, $(-1, 1)$ and $\left(\frac{1}{2}(-1 \pm \sqrt{5}), \frac{1}{2}(3 \mp \sqrt{5})\right)$; each satisfies $x = (y-2)^2 - 2$, e.g. $(2, 4)$: $(4-2)^2 - 2 = 2$.
- At the boundary $s = -\frac{3}{4}$ the quadratics are $x^2 - x - \frac{3}{4} = 0$ (roots $\frac{3}{2}, -\frac{1}{2}$) and $\left(x + \frac{1}{2}\right)^2 = 0$, so there are only two distinct common points, with $x = -\frac{1}{2}$ shared by both lines. A computer count of real roots of the quartic gave 4 for $s = -0.76, -1, -2$ and 2 for $s = -0.74, -0.5$.

Common slips

- Proving only the "if" direction in part (i): a pair of lines must give $p^2 - q^2 = 4r$, and you need an argument (such as the one above) for that.
- In part (iv), forgetting to check that the two lines don't share a point of C_2, or forgetting to show that the points found actually lie on C_1.

Beyond STEP

The family $F + kG = 0$ is a *pencil of conics*: every member passes through the common points of $F = 0$ and $G = 0$. Choosing k so that the member is degenerate (a pair of lines) is a classical way to find the intersections of two conics. In projective geometry this is done by making a 3×3 determinant vanish, which gives a cubic in k. Here the pencil happened to be degenerate for $k = 1$ whatever s is. Strang's *Introduction to Linear Algebra* covers the symmetric matrices behind conics. Siklos's *Advanced Problems in Mathematics* has more questions in this spirit.

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